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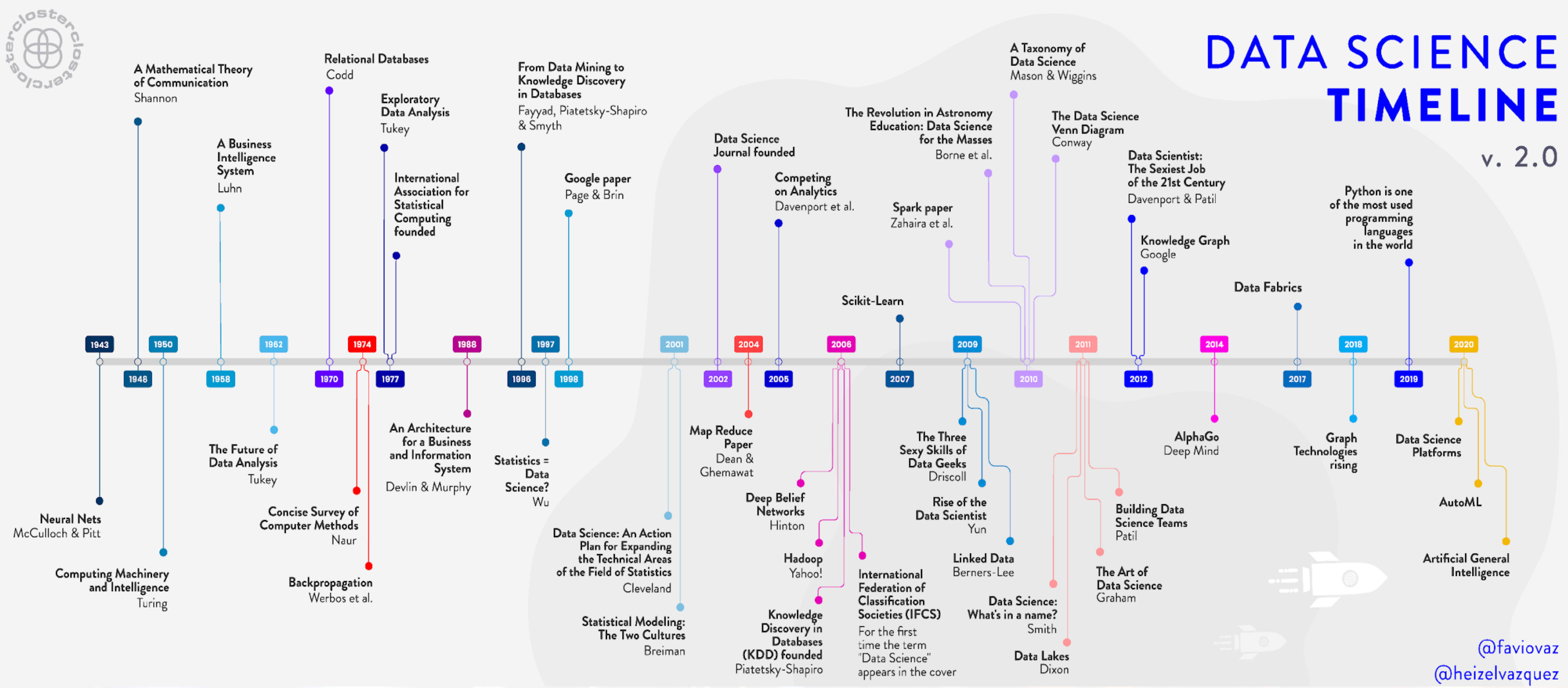
Introduction

Data Sciences : 06-July-2026

Data Science Timeline[1]



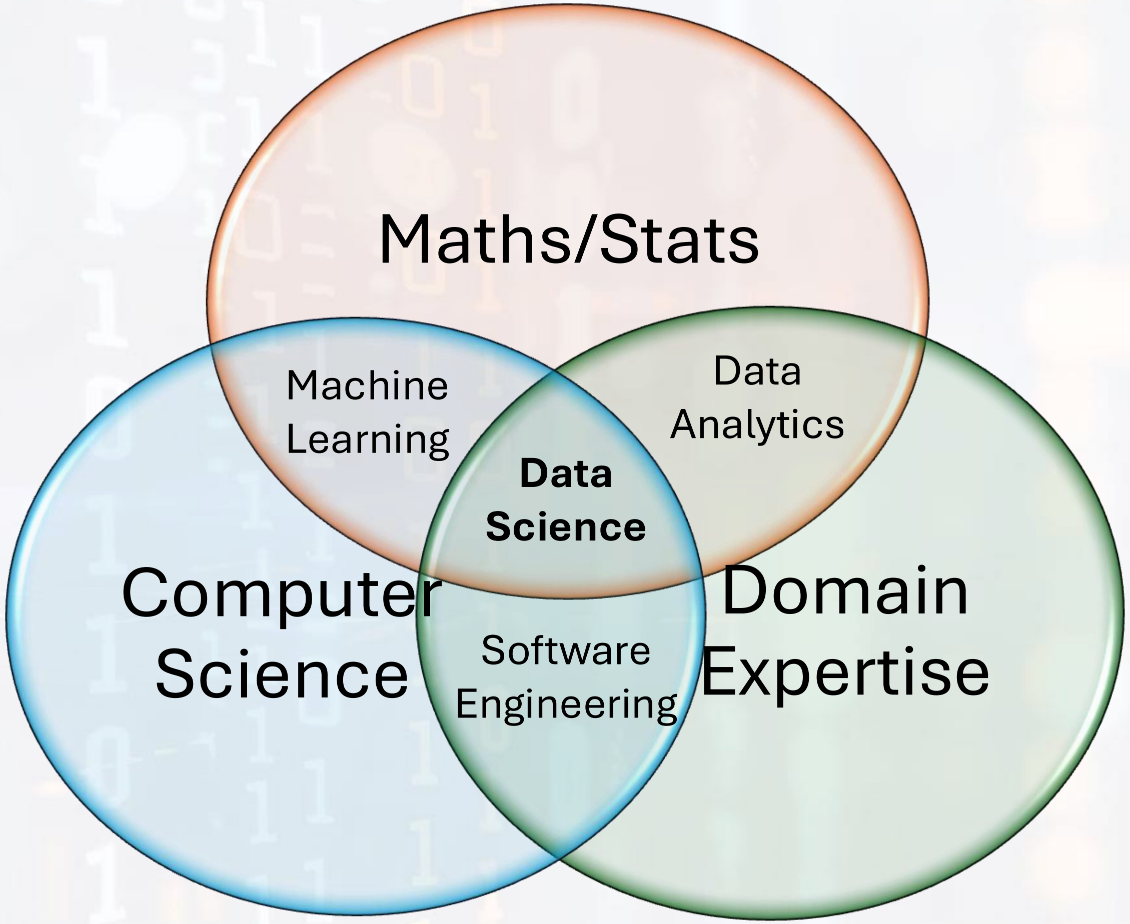
What is Data Science?

- There are many definitions of Data Science
- According to IBM[2]:
  - *"Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning"*

What is Data Science?

- *"Data science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data." [3]*
- *"… **extracts relevant insights** and **develops strategy from data** for business and industry. Combining tools, methods, and technology such as data analysis/modeling, human-machine interaction and algorithms, data scientists **ask and answer questions** like what happened, why did it happen, what will happen, and how can the results be extrapolated and used for planning and decision-making." [4]*

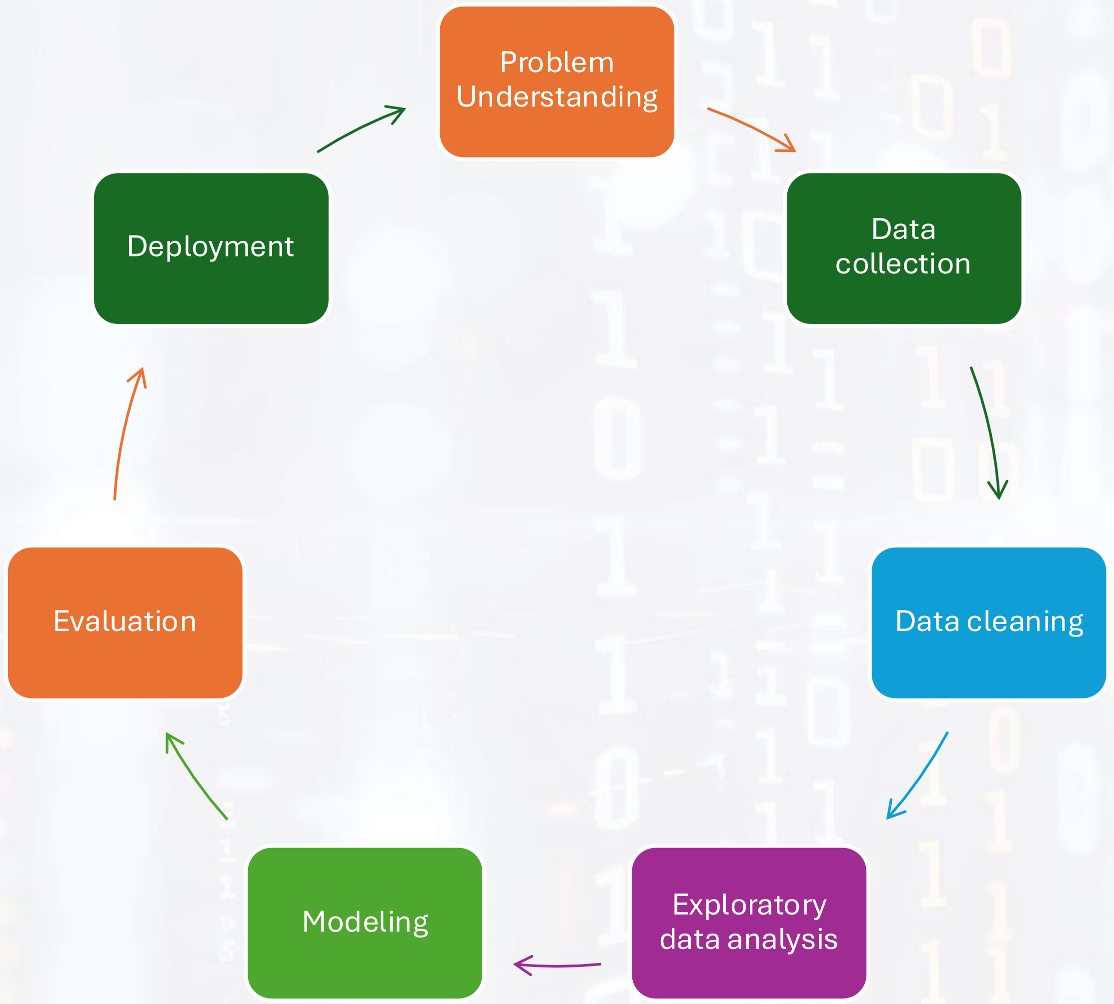
Data Science Components



Video on ‘How Data Science is shaping industry?’

- https://www.youtube.com/watch?v=nZREbKCNDCA

Data Science Life Cycle



Referenced from: [5]

- A day in the life of a data scientist – Applying the data science process

- Video link:
  - https://www.youtube.com/watch?v=X3paOmcrTjQ

Tools & Technologies

Commonly used:

- Languages: Python, R, SQL
- Libraries: pandas, scikit-learn, TensorFlow, matplotlib
- Platforms: Jupyter, Power BI, AWS, Azure

Skills needed by a Data Scientist [6]

Technical

Mathematics

Statistics

Computer science, Programming / Coding

Soft

Problem solving

Critical and analytical thinking

Curiosity

Communication

Team work

Types of Data Science Problems

Descriptive (what happened?)

- Summarize past data to understand what has occurred.
- Eg. Sales Reports: ‘Total sales by region last quarter’
- Healthcare: ‘Average patient wait time in hospitals last year’

Predictive (what will happen?)

- Use historical data to forecast future outcomes.
- Eg. Customer Churn Prediction ‘Which customers are likely to stop using our service?’
- Weather Forecasting: ‘Will it rain tomorrow?’

Prescriptive (what should we do?)

- Suggest actions or decisions based on predictions or simulations.
- Eg. Route Optimization: ‘What’s the best delivery route to minimize fuel costs?

Why is Data Science important?

Helps us make better decisions

Helps us identify problem and come up with a solution

Improves business processes

Get new insights

Predict the future

- Example: CSIR Elections predictions

The Future of Data Science

Trends: AI integration, AutoML, ethical data, usage, data governance, explainability.

Career outlook and job demand.

References

- [1] - https://medium.com/data-science/the-roots-of-data-science-77c71115229
- [2] - https://www.ibm.com/topics/data-science
- [3] - https://www.datacamp.com/blog/what-is-data-science-the-definitive-guide
- [4] - https://www.mtu.edu/data-science/what-is/
- [5] - https://www.sudeep.co/data-science/2018/02/09/Understanding-the-Data-Science-Lifecycle.html
- [6] - https://www.simplilearn.com/what-skills-do-i-need-to-become-a-data-scientist-article#nontechnical_skills_required_for_data_scientists

Questions?

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